

6. Assess systems and networks to identify security vulnerabilities, evaluate potential threats, and prioritize mitigation strategies based on risk
7. Deploy and configure biometric authentication methods to strengthen identity verification and enhance access control
8. Identify and mitigate Internet security risks including malware, phishing, and web-based attacks while securing wireless networks for mobile and remote access
9. Configure and manage cloud Identity and Access Management (IAM) solutions to control permissions, enforce least-privilege access, and protect cloud-based resources across distributed environments
10. Explain and implement zero-trust architecture principles—including continuous verification, least-privilege access, and micro-segmentation—to strengthen security in modern, perimeter-less networks
11. Deploy and manage Endpoint Detection and Response (EDR) platforms to monitor, detect, and remediate threats on endpoints using behavioral analysis and automated response capabilities
12. Use Open-Source Intelligence (OSINT) methodologies to gather, analyze, and validate publicly available information for identifying potential threats, vulnerabilities, or indicators of compromise

325 Network Design Team

Description & Eligibility

Teams will analyze existing and proposed business environments and develop a comprehensive strategy for implementing a network infrastructure that meets the organization's needs. This event evaluates the team's ability to assess requirements, plan scalable solutions, and present a clear, effective network design. Any secondary or postsecondary division contestant may enter this judged event.

A team will consist of 2-4 contestants. Teams must be registered for the national competition before the submission deadline and compete in both parts of the event to be ranked. Entries that do not follow the provided topic or the Copyright and/or Fair Use Guidelines and the Artificial Intelligence Usage & Citation Policy will be disqualified.

Topic

Cindershore Esports & Entertainment, headquartered in New York, New York, is a fast-growing organization that operates three flagship esports arenas inside major entertainment districts in Atlanta, Georgia; Denver, Colorado; and Phoenix, Arizona. Each arena hosts competitive gaming tournaments, live-streamed events, and public gaming lounges. Currently, each location relies on a patchwork of ISP-provided routers, unmanaged switches, and consumer-grade Wi-Fi access points, resulting in high latency during tournaments, inconsistent streaming quality, frequent network outages during peak events, and no centralized visibility or control across locations.

As part of their 2027 expansion, Cindershore plans to open three additional arenas in Charlotte, North Carolina; Sacramento, California; and Minneapolis, Minnesota. These new facilities will be purpose-built, approximately 8,000 square feet each, and will include a main tournament floor with 64 high-performance gaming stations, a broadcast studio for live streaming and commentary, a VIP lounge with premium internet access, administrative offices, and a public café area with complimentary Wi-Fi. Cindershore Esports & Entertainment needs a proposal for the following:

- **High-Throughput, Ultra-Low Latency Network Architecture:** Design a network capable of supporting simultaneous competitive gaming across 64 stations (minimum 25 Gbps backbone within each facility) with Quality-of-Service (QoS) prioritizing tournament traffic and keeping round-trip latency under 5ms on local segments.
- **Network Segmentation:** Create VLANs to isolate tournament gaming stations, broadcast/streaming equipment, VIP lounge, administrative offices, public café Wi-Fi, and building management/IoT systems.
- **Centralized Management and Monitoring:** Propose a unified network management solution (e.g., cloud-managed switches/APs or an SD-WAN controller) for all six sites, with real-time performance dashboards, automated alerting, and remote configuration capabilities.
- **Security and Remote Support:** Incorporate next-generation firewalls, intrusion detection/prevention systems (IDS/IPS), DDoS mitigation, VPN access for remote administration, and a role-based access control (RBAC) plan.
- **Scalability and Redundancy:** Ensure the design can support doubling tournament capacity and adding satellite pop-up event venues; include redundant ISP uplinks, failover strategies, and high-availability configurations for core switching.
- **Broadcast and Streaming Integration:** Plan dedicated network infrastructure supporting HD/4K live streaming from the broadcast studio, including bonded uplinks, CDN integration considerations, and stream isolation from public traffic.
- **Cost Estimation and ROI Analysis:** Provide a detailed bill of materials (switches, routers, APs, cabling, racks, licensing) and estimate both upfront capital expenditures and ongoing operational costs, balanced against projected revenue gains from improved tournament reliability and viewer experience.

Team Must Supply

- Published and/or unpublished non-electronic written reference materials
- Laptop/notebook computer (each team contestant may have one laptop), portable printer, software, paper and supporting equipment (extension cords, etc.)
- Students may use one single-sided laptop, notebook, or tablet for their presentation. Projection equipment and external displays will not be provided or permitted

Contest Notes

- Method of evaluation: judged
- AI Policy Level 2 *Permitted with Citation*
- Carry-in and setup of equipment must be done solely by the team and must take place within the allotted time for orientation/warm-up
- No Internet access will be provided on-site at NLC; however, contestants may provide their own access to be used only for their presentation to the judges
- Business Professionals of America assumes no responsibility for hardware/software provided by the Contestant
- No equipment, supplies or materials other than those specified for the event will be allowed in the testing area
- No personal electronic devices (smart watches, smart glasses, cell phone, etc.) – other than the laptop, notebook, or tablet approved for the presentation.
- No previous Business Professionals of America tests (handwritten, photocopied, or keyed) may be taken into the testing area. Violation of this rule will result in disqualification.
- Length of event: no more than three (3) minutes setup in presentation room, no more than ten (10) minutes presentation time, and no more than ten (10) minutes of judges' questions
 - No more than thirty (30) minutes preparation time at state and national level only
 - Teams will be presented with an additional element to the scenario that requires revision of their final presentation
- Each state is allowed three (3) entries

Technical Pre-Submit Specifications

What to Submit at https://upload.bpa.org	Saved File Name	National Deadline
Submit one combined PDF file: 1. Written Proposal 2. Works Cited 3. BPA Release Form(s) 4. AI Usage and Documentation Form	NDT-TeamID.pdf	April 1, 2027, 11:59 p.m. Eastern Time

Contest Competencies

1. Design, implement, and optimize network solutions, including routing strategies and VPNs, to meet organizational requirements for performance, security, and scalability
2. Design and document network cable topologies and create detailed network diagrams using industry-standard tools to support communication, troubleshooting, and future planning
3. Assess end-user requirements, plan resource placement, evaluate centralized or decentralized strategies, and incorporate scalability for future expansion in network designs
4. Design and implement redundancy and secure remote access solutions to ensure network availability and support distributed workforces
5. Evaluate project requirements, delegate responsibilities, and analyze technical support structures to ensure effective team collaboration and network maintenance
6. Apply analytical and problem-solving techniques to resolve complex networking challenges and optimize protocols, infrastructure, and host configurations

7. Assess organizational risk tolerance and develop multiple technical recommendations for networking scenarios to support informed decision-making
8. Prepare and present business reports that communicate technical findings, recommendations, and project outcomes to stakeholders
9. Design and evaluate SD-WAN architectures to improve application performance, enhance network reliability, and support secure connectivity across distributed enterprise environments
10. Incorporate zero-trust networking principles—including identity-based access, continuous verification, and micro-segmentation—into network designs to strengthen security and limit attack surfaces
11. Design wireless LAN controller (WLC) architectures and plan access point deployment strategies to ensure scalable, high-performance, and centrally managed wireless networks
12. Design and integrate cloud networking solutions by evaluating connectivity options, security models, and resource placement to support hybrid, multi-cloud, and cloud-native infrastructures

Contest Specifications for this Event

- The written proposal must follow the report format located in the *Style & Reference Manual*
- At the state and national levels, teams will be given an additional element to the original scenario to address during the presentation. These added elements are specific to the level of competition and are not connected to or carried over from previous competitions
- Cell phones may *not* be used in the preparation room
- No advisor contact will be allowed between the time of receiving the additional element and the delivery of the presentation
- Only team-prepared graphics may be used in the design. Originality and effectiveness will be given greater consideration

Common Specifications for all BPA Judged Events

- This is a pre-submitted event. Regional and State Conferences have specific deadlines prior to the National deadline. Please consult the regional or state conference administrator for regional or state deadlines. For the National Leadership Conference, submit the URL to <https://upload.bpa.org> no later than 11:59 p.m. Eastern Time, on April 1, 2027.
- No internet access will be provided on-site at the NLC. Contestants may bring their own internet access, but it may be used only during their presentation – not in the prep room.
- The team is responsible for securing a signed BPA Release Form(s) from any person, including themselves, whose image or work is used in the project.
- The team is responsible for citing all sources, including themselves, on the Works Cited page for any work used on the project. It is the policy of Business Professionals of America to comply with copyright law. In-text citations are required throughout the paper.
- The *Style & Reference Manual* contains guidelines for Copyright and/or Fair Use. Teams will be disqualified for violations of the guidelines.
- Materials must follow the organization's BPA Graphic Standards and make proper use of the BPA logo and/or organization's name. (Refer to the BPA Graphic Standards in the *Style & Reference Manual*.)
- Judges' comments will be returned digitally through the online judging system at the national level. Materials submitted for technical judging will *not* be returned and will *not* be available at NLC.

325 Network Design Team

Technical Scoring Rubric

Required Elements				
(If any question results in a NO, please assign a score of 0)				
Team followed the topic.				☐ Y ☐ N
Team followed Copyright and/or Fair Use Guidelines and the Artificial Intelligence Usage & Citation Policy.				☐ Y ☐ N
If answered no to either question, please stop scoring and provide a brief reason for the <i>disqualification</i> :				
Submission and Formatting				
(All points or none are awarded by the technical judge)				
Team submitted one (1) combined PDF file containing their Written Proposal, Works Cited formatted according to the <i>Style & Reference Manual</i> , BPA Release Form(s), and AI Usage and Documentation Form.				/10
Written Proposal	Below Expectations 0-6 points	Meets Expectations 7-13 points	Above Expectations 14-20 points	Points Awarded
Customer Profile	Incomplete or unclear; lacks relevance to topic.	Clear and relevant; basic customer needs identified.	Thorough and insightful; customer needs and context well-defined.	/20
Objectives	Vague or missing; lacks measurable goals.	Clear and measurable; aligned with topic.	Strategic and impactful; objectives show depth and foresight.	/20
Abstract of Implementation	Lacks clarity or detail; hard to follow.	Clear summary of implementation steps.	Exceptionally clear and well-structured; shows strong understanding.	/20
Explanation of Products/Services	Minimal or unclear explanation.	Adequate explanation with relevance to topic.	Detailed and compelling; clearly demonstrates value.	/20
Clarity of Message	Message is unclear or off-topic.	Message is clear, concise, and relevant	Message is compelling, creative, and perfectly aligned with topic.	/20
Short- and Long-Range Goals	Goals are missing or poorly defined	Goals are defined and realistic	Goals are strategic, ambitious, and well-supported.	/20
Financial Analysis	Missing or inaccurate; lacks justification.	Basic financials with reasonable justification.	Detailed, accurate, and insightful financial planning.	/20
Supporting Documentation	Missing or poorly organized.	Includes relevant and organized documentation.	Comprehensive, well-organized, and enhances understanding.	/20
Writing Accuracy	Frequent grammar or spelling errors.	Contains some grammar or spelling errors.	Contains very few or no grammar or spelling errors.	/20
Creativity	Below Expectations 0-6 Points	Meets Expectations 7-13 Points	Above Expectations 14-20 Points	Points Awarded
Innovation of Solution	Conventional or derivative ideas.	Some originality; meets basic expectations.	Highly innovative; fresh, unique, and creative approach.	/20

325 Network Design Team
Technical Scoring Rubric (Cont'd)

Specific Technical Recommendations	Below Expectations 0-10 Points	Meets Expectations 11-30 Points	Above Expectations 31-50 Points	Points Awarded
Design & Deployment Specifications	Missing or vague specifications.	Includes basic design and deployment details.	Detailed and technically sound; exceeds expectations.	/50
Specific Technical Recommendations	Below Expectations 0-6 Points	Meets Expectations 7-13 Points	Above Expectations 14-20 Points	Points Awarded
Network Structure & Setup	Incomplete or unclear setup.	Basic structure and setup provided.	Robust and scalable setup; clearly supports requirements.	/20
Bandwidth Tiers & Services	Missing or not well-defined.	Identifies basic tiers and services.	Clearly defined tiers; tailored to customer needs.	/20
Scalability & Quality of Service	No mention of scalability or QoS.	Addresses scalability and reliability.	Strong emphasis on scalability, QoS, and future-proofing.	/20
Value-Added Services	Not identified or irrelevant.	Identifies basic enhancements.	Creative and strategic enhancements that differentiate solution.	/20
TOTAL TECHNICAL POINTS				/340

325 Network Design Team

Presentation Scoring Rubric

Required Elements (If any question results in a NO, please assign a score of 0)				
Team followed the topic.				☐ Y ☐ N
Team followed Copyright and/or Fair Use Guidelines and Artificial Intelligence Usage & Citation Policy.				☐ Y ☐ N
If answered no to either question, please stop scoring and provide a brief reason for the <i>disqualification</i> :				
Evaluation Criteria	Below Expectations 0-5 points	Meets Expectations 6-10 points	Above Expectations 11-15 points	Points Awarded
Delivery & Engagement	Team rarely maintains eye contact, uses minimal gestures and body language, or has significant inconsistencies in tone and pace.	Team occasionally breaks eye contact, uses limited gestures and body language, or has some inconsistencies in tone and pace.	Team maintains good eye contact, uses appropriate gestures and body language, and varies their tone and pace adequately.	/15
Organization, Design & Structure	Presentation lacks coherence, and it is unclear how each team member's role contributes to the presentation. Transitions between speakers and/or points are awkward or absent.	Presentation content has some gaps, and it is not always clear how each team member's role contributes to the presentation. Transitions between speakers and/or points are choppy.	Presentation content flows adequately, and each team member's role is clear and integrated into the presentation. Transitions between speakers and/or points are smooth.	/15
Evaluation Criteria	Below Expectations 0-15 points	Meets Expectation 16-30 points	Above Expectations 31-45 points	Points Awarded
Content Knowledge	Presentation demonstrates a limited understanding of the topic and includes mostly irrelevant or inaccurate information.	Presentation demonstrates a basic understanding of the topic and includes a mix of relevant and irrelevant information.	Presentation demonstrates a good understanding of the topic and includes relevant and accurate information.	/45
Evaluation Criteria	Below Expectations 0-4 points	Meets Expectations 5-7 points	Above Expectations 8-10 points	Points Awarded
Judges' Questions	Responses lack depth and fail to demonstrate a comprehensive understanding of the topic.	Some responses lack detail or clarity. The understanding of the topic is partially demonstrated.	Responses are detailed, insightful, and effectively address the judges' questions.	/10
Team Members	At least two original team members were present during the presentation.			/5
Timing	Setup lasted no longer than three (3) minutes, and the presentation lasted no more than ten (10) minutes.			/10
TOTAL PRESENTATION POINTS				/100

PRESENTATION WILL BE STOPPED AT TEN (10) MINUTES